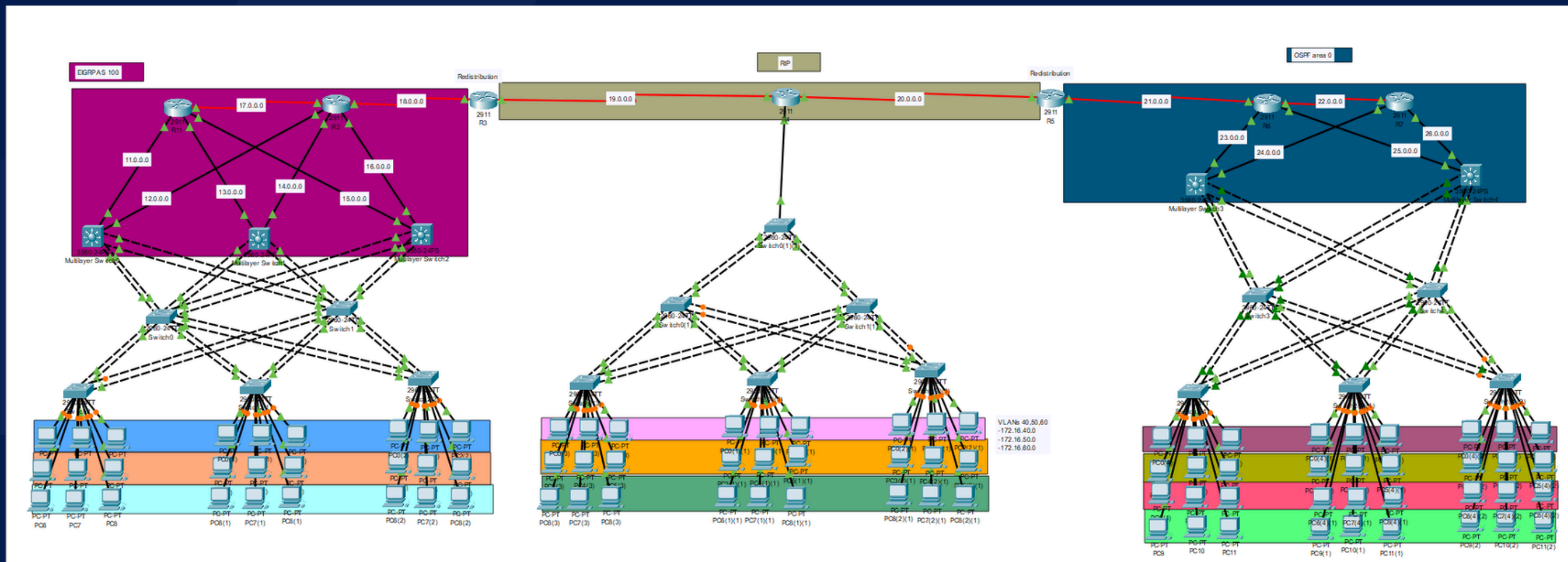




CCNA Project

presented by malak wael



A three-branch enterprise network implemented in Cisco Packet Tracer

Routing

Branch 1 — EIGRP

EIGRP AS 100

R1	interfaces	<pre>interface g0/0 ip address 11.0.0.2 255.255.255.252 ex interface g0/1 ip address 13.0.0.2 255.255.255.252 ex interface g0/2 ip address 15.0.0.2 255.255.255.252 ex interface s0/2/0 ip address 17.0.0.1 255.255.255.252 ex</pre>
	EIGRP	<pre>router eigrp 100 no-auto summary network 17.0.0.0 network 11.0.0.0 network 13.0.0.0 network 15.0.0.0</pre>

R2	interfaces	<pre>interface g0/0 ip address 12.0.0.2 255.255.255.252 ex interface g0/1 ip address 14.0.0.2 255.255.255.252 ex interface g0/2 ip address 16.0.0.2 255.255.255.252 ex interface s0/2/0 ip address 17.0.0.2 255.255.255.252 ex interface s0/2/1 ip address 18.0.0.1 255.255.255.252 ex</pre>
	EIGRP	<pre>router eigrp 100 no-auto summary network 17.0.0.0 network 18.0.0.0 network 12.0.0.0 network 14.0.0.0 network 16.0.0.0</pre>

Redistribution

EIGRP<-->RIP

R3	interfaces	<pre>interface s0/2/0 ip address 18.0.0.2 255.255.255.252 ex interface s0/2/1 ip address 19.0.0.1 255.255.255.252 ex</pre>
	EIGRP	<pre>router eigrp 100 no-auto summary network 18.0.0.0 redistribute rip metric 10000 100 255 1 1500</pre>
	RIP	<pre>router rip version 2 no-auto summary network 19.0.0.0 redistribute eigrp 100 metric 2</pre>

Branch 2 — RIP

RIP V2

R4	interfaces	<pre>interface s0/2/0 ip address 19.0.0.2 255.255.255.252 ex interface s0/2/1 ip address 12.0.0.1 255.255.255.252 ex</pre>
	RIP	<pre>router rip version 2 no-auto summary network 19.0.0.0 network 20.0.0.0</pre>

Redistribution

RIP<-->OSPF

R5	interfaces	<pre>interface s0/2/0 ip address 20.0.0.2 255.255.255.252 ex interface s0/2/1 ip address 21.0.0.1 255.255.255.252 ex</pre>
	RIP	<pre>router rip version 2 no-auto summary network 20.0.0.0 redistribute ospf 1 metric 2</pre>
	OSPF	<pre>router ospf 1 network 21.0.0.0 0.0.0.3 area 0 redistribute rip subnets</pre>

Branch 3 — OSPF

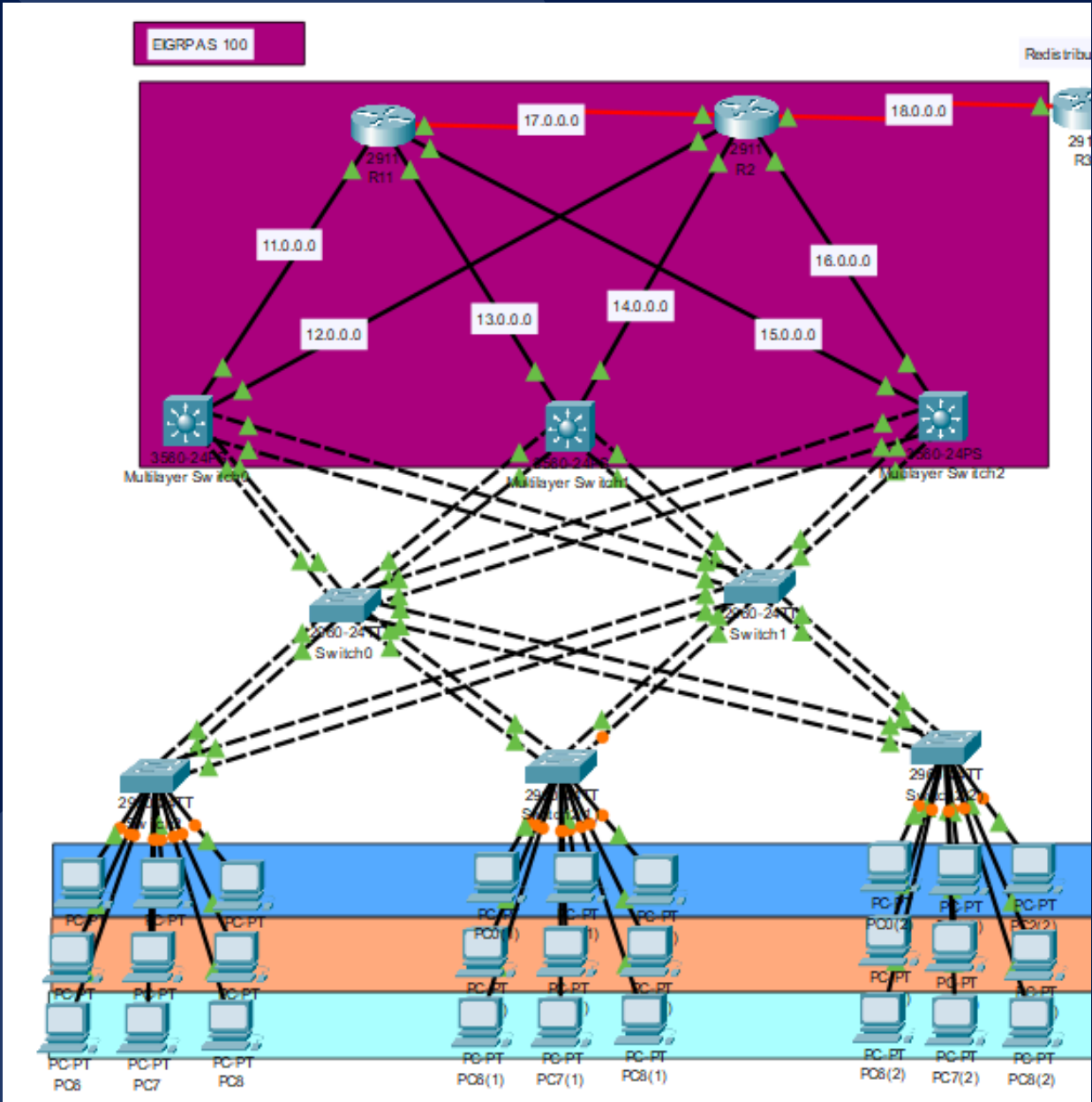
OSPF area 0

R6	interfaces	<pre>interface g0/0 ip address 23.0.0.2 255.255.255.252 ex interface g0/1 ip address 25.0.0.2 255.255.255.252 ex interface s0/2/0 ip address 21.0.0.2 255.255.255.252 ex interface s0/2/1 ip address 22.0.0.1 255.255.255.252 ex</pre>
	OSPF	<pre>router ospf 1 network 21.0.0.0 0.0.0.3 area 0 network 22.0.0.0 0.0.0.3 area 0 network 23.0.0.0 0.0.0.3 area 0 network 25.0.0.0 0.0.0.3 area 0</pre>

R7	interfaces	<pre>interface g0/0 ip address 24.0.0.2 255.255.255.252 ex interface g0/1 ip address 26.0.0.2 255.255.255.252 ex interface s0/2/0 ip address 22.0.0.2 255.255.255.252 ex</pre>
	OSPF	<pre>router ospf 1 network 22.0.0.0 0.0.0.3 area 0 network 24.0.0.0 0.0.0.3 area 0 network 26.0.0.0 0.0.0.3 area 0</pre>

Switching

Branch 1



On Multilayer switch

Trunks	<pre>interface range f0/1-4 switchport trunk encapsulation dot1q switchport mode trunk switchport trunk allowed vlan all</pre>
Etherchannel	<pre>interface range f0/1-2 channel-group 1 mode active ex interface range f0/3-4 channel-group 2 mode active</pre>
Vlans	<pre>vlan 10 name it ex vlan 20 name hr ex vlan 30 name dev ex</pre>
VTP	<pre>vtp mode server vtp domain branch1 vtp password branch1</pre>
STP/RSTP	<pre>spanning-tree vlan 10 priority 4096 spanning-tree vlan 20,30 priority 8192 spanning-tree mode rapid-pvst</pre>

SVI & DHCP

MLS1

SVI	<pre>ip routing interface vlan 10 ip address 192.168.10.1 255.255.255.0 no shutdown standby 10 ip 192.168.10.50 standby 10 priority 150 standby 10 preempt interface vlan 20 ip address 192.168.20.1 255.255.255.0 no shutdown standby 10 ip 192.168.20.50 standby 10 priority 100 interface vlan 30 ip address 192.168.30.1 255.255.255.0 no shutdown standby 10 ip 192.168.30.50 standby 10 priority 100 interface f0/5 no switchport ip address 11.0.0.1 255.255.255.252 interface f0/6 no switchport ip address 12.0.0.1 255.255.255.252</pre>
	<pre>ip dhcp pool vlan10 network 192.168.10.0 255.255.255.0 default-router 192.168.10.50 dns-server 8.8.8.8 ex ip dhcp excluded-address 192.168.10.1 192.168.10.3 ip dhcp excluded-address 192.168.10.50</pre>

MLS2

SVI	<pre>ip routing interface vlan 10 ip address 192.168.10.2 255.255.255.0 no shutdown standby 10 ip 192.168.10.50 standby 10 priority 100 interface vlan 20 ip address 192.168.20.2 255.255.255.0 no shutdown standby 10 ip 192.168.20.50 standby 10 priority 150 standby 10 preempt interface vlan 30 ip address 192.168.30.2 255.255.255.0 no shutdown standby 10 ip 192.168.30.50 standby 10 priority 100 interface f0/5 no switchport ip address 13.0.0.1 255.255.255.252 interface f0/6 no switchport ip address 14.0.0.1 255.255.255.252</pre>
	<pre>ip dhcp pool vlan20 network 192.168.20.0 255.255.255.0 default-router 192.168.20.50 dns-server 8.8.8.8 ex ip dhcp excluded address 192.168.20.1 192.168.20.3 ip dhcp excluded address 192.168.20.50</pre>

MLS3

SVI	<pre>ip routing interface vlan 10 ip address 192.168.10.3 255.255.255.0 no shutdown standby 10 ip 192.168.10.50 standby 10 priority 100 interface vlan 20 ip address 192.168.20.3 255.255.255.0 no shutdown standby 10 ip 192.168.20.50 standby 10 priority 100 interface vlan 30 ip address 192.168.30.3 255.255.255.0 no shutdown standby 10 ip 192.168.30.50 standby 10 priority 150 standby 10 preempt interface f0/5 no switchport ip address 15.0.0.1 255.255.255.252 interface f0/6 no switchport ip address 16.0.0.1 255.255.255.252</pre>
	<pre>ip dhcp pool vlan30 network 192.168.30.0 255.255.255.0 default-router 192.168.30.50 dns-server 8.8.8.8 ex ip dhcp excluded address 192.168.30.1 192.168.30.3 ip dhcp excluded address 192.168.30.50</pre>

Distribute switch

Trunks	interface range f0/1-12 switchport mode trunk switchport trunk allowed vlan all
Etherchannel	interface range f0/1-2 channel-group 1 mode active ex interface range f0/3-4 channel-group 2 mode active ex interface range f0/5-6 channel-group 3 mode active ex interface range f0/7-8 channel-group 4 mode passive ex interface range f0/9-10 channel-group 5 mode passive ex interface range f0/11-12 channel-group 6 mode passive ex
VTP	vtp mode client vtp domain branch1 vtp password branch1
STP/RSTP	spanning-tree vlan 10,20,30 priority 12288 spanning-tree mode rapi-pvst

Access switch

Trunks	interface range f0/10-13 switchport mode trunk switchport trunk allowed vlan all
Etherchannel	interface range f0/10-11 channel-group 1 mode passive ex interface range f0/12-13 channel-group 2 mode passive ex
VTP	vtp mode client vtp domain branch1 vtp password branch1
STP/RSTP	spanning-tree vlan 10,20,30 priority 16384 spanning-tree mode rapi-pvst
Access ports/STP features	interface range f0/1-9 switchport mode access spanning-tree portfast spanning-tree bpduguard enable switchport port-security switchport port-security maximum 1 switchport port-security mac-address sticky switchport port-security violation shutdown ex interface range f0/1-3 switchport access vlan 10 ex interface range f0/4-6 switchport access vlan 20 ex interface range f0/7-9 switchport access vlan 30 ex

Branch 2

Router	sub-interface	<pre>interface g0/0/0.40 encapsulation dot1q 40 ip address 172.16.40.1 255.255.255.0 ex interface g0/0/0.50 encapsulation dot1q 50 ip address 172.16.50.1 255.255.255.0 ex interface g0/0/0.60 encapsulation dot1q 60 ip address 172.16.60.1 255.255.255.0</pre>
	DHCP	<pre>ip dhcp pool vlan40 network 172.16.40.0 255.255.255.0 default-router 192.168.40.1 dns-server 8.8.8.8 ex ip dhcp excluded address 172.16.40.1 ip dhcp pool vlan50 network 172.16.50.0 255.255.255.0 default-router 192.168.50.1 dns-server 8.8.8.8 ex ip dhcp excluded address 172.16.50.1 ip dhcp pool vlan60 network 172.16.60.0 255.255.255.0 default-router 192.168.60.1 dns-server 8.8.8.8 ex ip dhcp excluded address 172.16.60.1</pre>

Core	Trunks	<pre>interface range f0/1-5 switchport mode trunk switchport allowed vlan all</pre>
	Etherchannel	<pre>interface range f0/1-2 channel-group 1 mode active ex interface range f0/3-4 channel-group 2 mode active</pre>
	Vlans	<pre>vlan 40 name it ex vlan 50 name hr ex vlan 60 name dev ex</pre>
	VTP	<pre>vtp mode server vtp domain branch2 vtp password branch2 ex</pre>
	STP/RSTP	<pre>spanning-tree vlan 40,50,60 priority 4096 spanning-tree mode rapid-pvst</pre>

DIS	Trunks	interface range f0/1-8 switchport mode trunk switchport allowed vlan all
	Etherchannel	interface range f0/7-8 channel-group 1 mode passive ex interface range f0/1-2 channel-group 2 mode active ex interface range f0/3-4 channel-group 3 mode active ex interface range f0/5-6 channel-group 4 mode active ex
	VTP	vtp mode client vtp domain branch2 vtp password branch2 ex
	STP/RSTP	spanning-tree vlan 40,50,60 priority 8192 spanning-tree mode rapid-pvst

Access	Trunk	interface range f0/10-13 switchport mode trunk switchport allowed vlan all
	Etherchannel	interface range f0/10-11 channel -group 1 mode passive ex interface range f0/12-13 channel -group 2 mode passive ex
	VTP	vtp mode client vtp domain branch2 vtp password branch2 ex
	STP/RSTP	spanning-tree vlan 40,50,60 priority 12288 spanning-tree mode rapid-pvst
	Access Ports /STP feature	interface range f0/1-9 switchport mode access spanning-tree portfast spanning-tree bpduguard enable switchport port-security switchport port-security maximum 1 switchport port-security mac-address sticky switchport port-security violation shutdown ex interface range f0/1-3 switchport access vlan 40 ex interface range f0/4-6 switchport access vlan 50 ex interface range f0/7-9 switchport access vlan 60 ex

Branch 3

On Multilayer switch

Trunks	<pre>interface range f0/1-4 switchport trunk encapsulation dot1q switchport mode trunk switchport trunk allowed vlan all</pre>
Etherchannel	<pre>interface range f0/1-2 channel-group 1 mode active ex interface range f0/3-4 channel-group 2 mode active</pre>
Vlans	<pre>vlan 70 name lt ex vlan 80 name hr ex vlan 90 name dev ex vlan 100 name cs ex</pre>
VTP	<pre>vtp mode server vtp domain branch3 vtp password branch3</pre>
STP/RSTP	<pre>spanning-tree vlan 70,80 priority 4096 spanning-tree vlan 90 priority 8192 spanning-tree mode rapid-pvst</pre>

MLS1

	<pre>ip routing interface vlan 70 ip address 10.0.70.1 255.255.255.0 no shutdown standby 10 ip 10.0.70.50 standby 10 priority 150 standby 10 preempt interface vlan 80 ip address 10.0.80.1 255.255.255.0 no shutdown standby 10 ip 10.0.80.50 standby 10 priority 150 standby 10 preempt interface vlan 90 ip address 10.0.90.1 255.255.255.0 no shutdown standby 10 ip 10.0.90.50 standby 10 priority 100 interface vlan 100 ip address 10.0.100.1 255.255.255.0 no shutdown standby 10 ip 10.0.100.50 standby 10 priority 100 interface f0/5 no switchport ip address 23.0.0.1 255.255.255.252 interface f0/6 no switchport ip address 24.0.0.1 255.255.255.252</pre>
SVI	
DHCP	<pre>ip dhcp pool vlan70 network 10.0.70.0 255.255.255.0 default-router 10.0.70.50 dns-server 8.8.8.8 ex ip dhcp excluded address 10.0.70.1 ip dhcp excluded address 10.0.70.2 ip dhcp excluded address 10.0.70.50 ip dhcp pool vlan80 network 10.0.80.0 255.255.255.0 default-router 10.0.80.50 dns-server 8.8.8.8 ex ip dhcp excluded address 10.0.80.1 ip dhcp excluded address 10.0.80.2 ip dhcp excluded address 10.0.80.50</pre>

MLS2

	<pre>ip routing interface vlan 70 ip address 10.0.70.2 255.255.255.0 no shutdown standby 10 ip 10.0.70.50 standby 10 priority 100 interface vlan 80 ip address 10.0.80.2 255.255.255.0 no shutdown standby 10 ip 10.0.80.50 standby 10 priority 100 interface vlan 90 ip address 10.0.90.2 255.255.255.0 no shutdown standby 10 ip 10.0.90.50 standby 10 priority 150 standby 10 preempt interface vlan 100 ip address 10.0.100.2 255.255.255.0 no shutdown standby 10 ip 10.0.100.50 standby 10 priority 150 standby 10 preempt interface f0/5 no switchport ip address 26.0.0.1 255.255.255.252 interface f0/6 no switchport ip address 25.0.0.1 255.255.255.252</pre>
SVI	
DHCP	<pre>ip dhcp pool vlan90 network 10.0.90.0 255.255.255.0 default-router 10.0.90.50 dns-server 8.8.8.8 ex ip dhcp excluded address 10.0.90.1 ip dhcp excluded address 10.0.90.2 ip dhcp excluded address 10.0.90.50 ip dhcp pool vlan100 network 10.0.100.0 255.255.255.0 default-router 10.0.100.50 dns-server 8.8.8.8 ex ip dhcp excluded address 10.0.100.1 ip dhcp excluded address 10.0.100.2 ip dhcp excluded address 10.0.100.50</pre>

Distribute switch

Trunks	interface range f0/1-10 switchport mode trunk switchport trunk allowed vlan all
Etherchannel	interface range f0/1-2 channel-group 1 mode active ex interface range f0/3-4 channel-group 2 mode active ex interface range f0/5-6 channel-group 3 mode active ex interface range f0/7-8 channel-group 4 mode passive ex interface range f0/9-10 channel-group 5 mode passive ex
VTP	vtp mode client vtp domain branch3 vtp password branch3
STP/RSTP	spanning-tree vlan 70,80,90 priority 12288 spanning-tree mode rapi-pvst

Access switch

Trunks	interface range f0/13-16 switchport mode trunk switchport trunk allowed vlan all
Etherchannel	interface range f0/13-14 channel-group 1 mode passive ex interface range f0/15-16 channel-group 2 mode passive ex
VTP	vtp mode client vtp domain branch3 vtp password branch3
STP/RSTP	spanning-tree vlan 70,80,90 priority 16384 spanning-tree mode rapi-pvst
Access ports/STP features	interface range f0/1-12 switchport mode access spanning-tree portfast spanning-tree bpduguard enable switchport port-security switchport port-security maximum 1 switchport port-security mac-address sticky switchport port-security violation shutdown ex interface range f0/1-3 switchport access vlan 70 ex interface range f0/4-6 switchport access vlan 80 ex interface range f0/7-9 switchport access vlan 90 ex interface range f0/10-12 switchport access vlan 100 ex